Chapter 3
Hedging Strategies Using Futures

Most participants in futures markets are there to reduce a risk they already have, not to acquire one. This chapter is about how they do it and, more usefully, about the ways in which it does not quite work.

Futures are only one way to do it, and it is worth being explicit about why a firm chooses them. They are *liquid*, so a large hedge can be put on and taken off the same day. They are *cheap*, with spreads a fraction of the equivalent bank quote. They carry *no bilateral exposure to the other side of the trade*, because the clearing house of Chapter 2 stands between the two sides (exposure to the clearing house and to the clearing broker remains, as Chapter 2 explains). And they cost *nothing at inception*, which matters more to a treasurer defending a budget than it should. The alternatives each give one of these up: a forward matches the exposure exactly but is illiquid and carries the dealer's credit, a swap does the same across many dates, and an option keeps the upside but has to be paid for.

A hedge that worked perfectly would be uninteresting. The producer would lock in a price, the consumer would lock in a cost, and there would be nothing further to say. In practice four things intervene: the futures contract may not match the exposure, the hedge may be lifted before the contract expires, the exposure may last longer than any liquid contract, and (the theme Chapter 2 introduced and Chapter 8 develops) the hedge generates cash flows on a schedule the hedged item does not. *Each is the shadow of one of the advantages just listed.* Standardization is what makes futures liquid and what makes them fit imperfectly; a small number of listed maturities is what concentrates the liquidity and what forces the roll; daily settlement is what removes the bilateral credit risk and what demands the cash. A hedger does not get to take the advantages without their costs, and choosing futures is choosing that bundle.

Each of these is a source of residual risk, and each has cost real firms large sums. Sections 3.3 through 3.9 take them in turn, and Section 3.10 examines what happens when they combine.

3.1 SHORT AND LONG HEDGES

A **short hedge** is appropriate when a firm already owns an asset, or expects to own one, and wants to protect against a fall in its price. The firm sells futures. If the price falls, the loss on the physical position is offset by a gain on the futures.

A **long hedge** is appropriate when a firm knows it will need to buy an asset in future and wants to protect against a rise in price. The firm buys futures.

Example 3.1 An oil producer's short hedge

On May 15 a producer contracts to sell 1 million barrels of crude on August 15 at whatever the spot price is on that day. Spot is currently \$80 and the September futures price is \$80.50. Each contract covers 1,000 barrels, so the producer sells 1,000 contracts.

On August 15 the spot price has fallen to \$72 and the September futures price is \$72.10.

Revenue from selling the oil	\$72,000,000
Gain on the futures position, $(80.50 - 72.10) \times 1{,}000{,}000$	\$8,400,000
Total	\$80,400,000

The effective price received is \$80.40 per barrel: close to, but not exactly, the \$80.50 futures price at which the hedge was placed.

Note the discrepancy of 10 cents. It is not an error, and it is the subject of Section 3.3.

3.2 ARGUMENTS FOR AND AGAINST HEDGING

Before the mechanics comes a question that matters more than it is usually given credit for: should a firm hedge at all?

The case against. Shareholders can diversify. If an investor holds an oil producer alongside an airline, the oil price risk partly cancels within the portfolio, and the investor did not need either company to hedge. This is the Modigliani–Miller argument applied to risk: in frictionless markets, whatever a firm does to its risk its shareholders could do or undo on their own account, so hedging cannot add value. Hedging costs money and management attention, and a firm that hedges away the exposure its shareholders bought it for may be destroying the thing they wanted. Gold mining is the clearest case. Producers sold their output forward through the 1990s, and when the gold price rose through the decade that followed, shareholders who had bought the miners precisely to own gold discovered the exposure had been sold out from under them. The industry then spent years, and a great deal of money, buying those hedge books back.

There is also a competitive argument. If an entire industry faces the same input cost and passes it through, hedging does not reduce a firm's risk relative to its competitors—it *creates* it, because the hedged firm's costs now diverge from everyone else's. Airlines are the standing example: when fuel is expensive every carrier raises fares together, so the hedged carrier gains only until prices fall, at which point it is the one carrier that cannot cut.

The case for. The strongest arguments are about frictions the diversification story ignores. *Financial distress is costly*: a firm that runs out of cash loses customers, suppliers, and staff in ways that a shareholder's diversified portfolio does not compensate for, so reducing the probability of that state has value even if the expected outcome is unchanged. Lenders act on this directly: a reserve-based loan to an oil and gas producer, whose size (the **borrowing base**) is set by the value of the producer's reserves, commonly *requires* that a share of production be hedged as a condition of that borrowing base, because what the lender is buying is the reduced probability of distress and not a view on the oil price. Froot, Scharfstein, and Stein (1993) sharpen the point: when outside finance costs more than internal cash, a shortfall in cash flow forces a firm to pass up good investments, so a hedge that keeps internal funds available in bad states protects investment, not merely earnings. Taxes argue the same way, as Smith and Stulz (1985) show: where the tax schedule is convex, as progressive rates and limits on the use of losses make it, a smoother taxable income lowers the expected tax bill. *Managers know more than shareholders* about the firm's exposures, and can hedge more cheaply than shareholders can replicate. And *planning is easier* when input costs are known: a firm that can commit to a price can sign contracts it otherwise could not. An airline sells tickets months before it buys the fuel to fly them, and a utility files tariffs it cannot revisit for a year; neither can quote a price today without knowing a cost it will not pay until later.

Hedging is worth doing when it reduces the probability of genuinely bad states, and is closer to a wash when it merely smooths outcomes within a range the firm can absorb.

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Case Study 3.1 The hedger who is punished for hedging

A hedge that works generates a loss on one leg. That is what a hedge is.

The difficulty is that the two legs are usually not reported the same way. The futures loss is realized, visible, and appears in a specific line of the accounts with someone's name against it. The offsetting gain sits in an operating result, unlabeled, and looks like ordinary business performance. When prices move against the futures leg, the person who put the hedge on has a demonstrable loss to explain, and the person who did nothing has a good year.

Airlines have provided the clearest examples in both directions. Carriers that hedged fuel were criticized for the cost when prices fell, and praised when prices rose; carriers that did not hedge faced the reverse. The underlying decision, whether to accept a known cost or an unknown one, was the same in every year, but it was judged each time by an outcome that the decision was explicitly designed not to depend on.

The practical consequence is that hedging programs need to be justified and measured on the combined position, before the fact, with the mandate agreed in advance. A program evaluated leg by leg after the fact will be abandoned at the worst possible moment. That is what Section 3.10 describes happening to a real firm.

3.3 BASIS RISK

Example 3.1 did not lock in $80.50. The reason is that the hedge involved two prices (the spot price the producer received and the futures price at which the hedge was closed), and they need not move together.

The **basis** is defined as

$$\text{basis} = \text{spot price of the asset to be hedged} - \text{futures price of the contract used} \tag{3.1}$$

Chapter 2 showed the basis converges to zero at delivery. Before delivery it does not, and it varies.

Write S_1, F_1 for the spot and futures prices when the hedge is placed, S_2, F_2 for their values when it is lifted, and $b_1 = S_1 - F_1$, $b_2 = S_2 - F_2$ for the corresponding bases. For a short hedge the effective price received is

$$S_2 + (F_1 - F_2) = F_1 + b_2 \tag{3.2}$$

and the identical expression holds for a long hedge, as the effective price paid.

Equation (3.2) is the central result of this chapter. The hedger locks in F_1, which is known today, plus the basis at the time the hedge is closed, which is not. All the residual uncertainty is in b_2.

Figure 3.1 shows the two prices converging. The gap narrows to zero at delivery, but a hedger who lifts the hedge before then is exposed to wherever the gap happens to be at that moment.

The figure is also equation (3.2) drawn. The hedger sells the physical at S_2 and closes the futures for $F_1 - F_2$, so S_2 plus the futures gain $F_1 - F_2$ reaches the dashed horizontal line above the two marked points, and that line sits above F_1 by exactly the gap b_2 between them. Both readings give the same height, which is what the identity says. Notice what has dropped out: the sinuous common movement of the two paths appears in S_2 and in F_2 and cancels between them, so the effective price does not depend on where prices went, only on the gap that remained.

Who gains from which move. An increase in the basis is called a **strengthening**; a decrease is a **weakening**. From (3.2), a short hedger receives $F_1 + b_2$ and therefore **benefits from**

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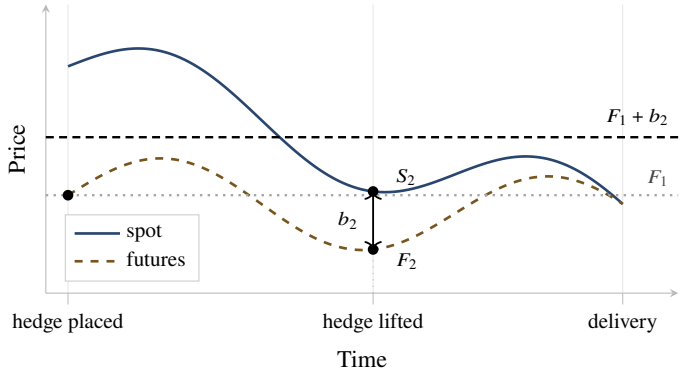


Figure 3.1: The basis closes at delivery; what survives at the lift date is b_2. Schematic.

an unexpected strengthening. A long hedger pays the same quantity and benefits from a weakening. Example 3.2 puts numbers on a weakening.

Example 3.2 Basis risk in numbers

A hedge is placed when the spot price is \$62 and the futures price is \$60, so $b_1 = +\$2$. It is lifted when the spot price is \$58 and the futures price is \$57.50, so $b_2 = +\$0.50$.

For a short hedger the effective price is

$$F_1 + b_2 = 60 + 0.50 = \$60.50$$

Had the basis held at \$2, the effective price would have been \$62. The basis weakened by \$1.50, and the short hedger lost exactly that amount per unit relative to expectation—despite the hedge having done its job in every other respect.

Compare the unhedged outcome. The spot price fell \$4 from the \$62 at which the hedge was placed. The hedge absorbed \$2.50 of the fall and left \$1.50, and that residual is basis risk.

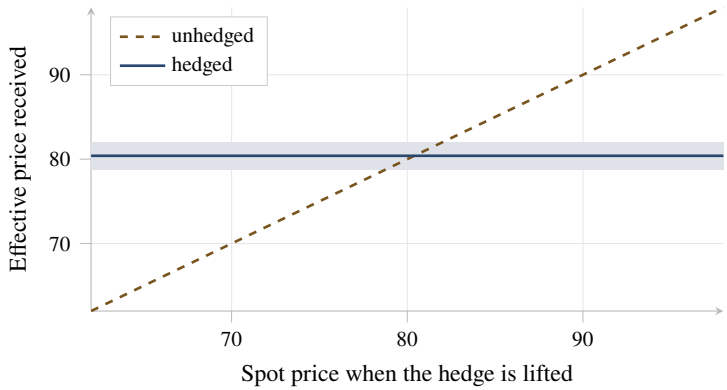


Figure 3.2: Effective price received against the spot price at the time the hedge is lifted. The band is basis risk.

Figure 3.2 makes the point visually. Unhedged, the effective price is whatever the spot price turns out to be (the diagonal). Hedged, it is $F_1 + b_2$, which is flat in the spot price: the hedge has removed the exposure entirely. What remains is the vertical thickness of the band, which is the variability of b_2.

The figure also shows what the hedger has given up. Everything to the right of the crossing

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point is upside the hedger will not receive, and every hedging program has to survive a year in which that region is where the market went. This is the organizational problem of Case Study 3.1, drawn.

Where basis risk comes from. Basis risk has three sources, each corresponding to a mismatch:

- **Asset mismatch.** The asset hedged differs from the deliverable grade: jet fuel hedged with heating oil (since 2013 the New York Mercantile Exchange (NYMEX) contract of that name has specified ultra-low-sulfur diesel delivered in New York Harbor), a corporate bond portfolio hedged with Treasury futures. Chapter 2's remark that the deliverable grade determines what a contract tracks is the same point.
- **Location mismatch.** The same commodity at a different delivery point, connected only by transport capacity that can bind (the reason Chapter 2 flagged delivery location as economically significant).
- **Timing mismatch.** The hedge is lifted before the contract expires, so convergence is incomplete.

Choosing the contract. Two decisions reduce basis risk. Choose the asset whose price is most closely correlated with the exposure, and choose a delivery month *after* the hedge will be lifted, but not much after, since liquidity concentrates in near contracts. Hedging a June exposure with a July contract is usually better than with a June contract, since holding into the delivery month exposes the hedger to delivery mechanics and to erratic prices.

3.4 BASIS RISK BEYOND COMMODITIES

Basis risk is usually introduced with commodities, where the mismatches are physical and easy to picture. It is at least as important in financial hedges, where it is less visible and therefore more dangerous.

Credit. A portfolio of corporate bonds hedged with Treasury futures is protected against the level of interest rates and not at all against credit spreads. In calm markets the two move together enough that the hedge looks effective; in a credit sell-off, government yields *fall* while corporate yields rise, so both legs lose simultaneously. The hedge's measured R^2 over a quiet sample (the share of the exposure's variance the hedge removed, which Section 3.5 defines) says nothing about this, because the sample contains no such episode.

This is the general form of the problem, and it shapes how the next section should be read: a hedge ratio estimated over a period in which the basis behaved will fail in the period when the basis is what moves.

Currency. A firm with an exposure in a currency that has no liquid futures contract hedges with a correlated one: a Norwegian krone receivable hedged with euro futures, say. The hedge works while the two currencies track and fails when they do not, and they most reliably fail to track when something specific is happening to one of them, which is when the exposure matters.

Rates. Chapter 7's swap spread is a basis. A dealer hedging a swap book with Treasury futures, as Chapter 6 describes, is exposed to the spread between the swap curve and the government curve, and Chapter 7 shows that spread moving to a place no one expected it to go.

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The common structure. In each case a hedger has substituted a liquid instrument for the exposure it actually holds and retained the difference. The difference is typically small, mean-reverting, and well behaved—until it is none of those things. A hedger should be able to state, before placing a hedge, what the basis is, what would make it move, and how much it could move; a hedger who can only state the historical correlation has not answered the question.

3.5 CROSS HEDGING AND THE MINIMUM VARIANCE HEDGE RATIO

When the hedged asset differs from the futures asset (a **cross hedge**), a one-for-one position is generally not optimal. If the futures price is more volatile than the spot price, a smaller futures position gives a better hedge.

The **hedge ratio** h is the size of the futures position relative to the exposure. The value minimizing the variance of the hedged position, the **minimum-variance hedge ratio**, is

$$\boxed{h^* = \rho \frac{\sigma_S}{\sigma_F}} \tag{3.3}$$

where σ_S and σ_F are the standard deviations of changes in the spot and futures prices over the hedge's life, and ρ their correlation. [TN3] derives it, extends it to several futures contracts at once, and shows what the ratio costs when its three inputs are estimated rather than known. That cost decides whether a second hedging contract is worth using at all.

If the two prices move identically, $\rho = 1$ and $\sigma_S = \sigma_F$, giving $h^* = 1$: hedge one-for-one. Any imperfection moves the optimal ratio away from one. A correlation below one always moves it downward, relative to a perfectly correlated hedge with the same volatility ratio, because a futures position that does not reliably offset should be smaller. A volatility mismatch can move it either way: above one when the spot price is the more volatile, below one when the futures price is.

The connection to regression. Equation (3.3) is the slope coefficient from regressing changes in the spot price on changes in the futures price. This is not a coincidence: minimizing squared deviations is what both problems do. It is also how h^* is estimated in practice. The regression's R^2 then measures the proportion of variance the hedge removes.

The optimal number of contracts. With Q_A units of exposure and Q_F units per contract,

$$N^* = \frac{h^* Q_A}{Q_F} \tag{3.4}$$

Example 3.3 applies both formulas.

Example 3.3 Hedging jet fuel with heating oil

An airline expects to buy 2 million gallons of jet fuel in one month. There is no liquid jet fuel futures contract, so it uses heating oil futures, each covering 42,000 gallons, as a cross hedge.

From historical data on monthly changes: the standard deviation of the jet fuel price change is \$0.032 per gallon, that of the heating oil futures price change is \$0.040, and their correlation is 0.8. From (3.3),

$$h^* = 0.8 \times \frac{0.032}{0.040} = 0.64$$

and from (3.4),

$$N^* = \frac{0.64 \times 2{,}000{,}000}{42{,}000} \approx 30.5$$

so the airline buys 30 contracts.

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Two observations follow. The hedge ratio is well below one (the airline's 30 contracts cover 1.26 million gallons against a 2 million gallon exposure) because heating oil futures are more volatile than jet fuel and imperfectly correlated with it. And the hedge is far from perfect: with $\rho = 0.8$, the regression R^2 is 0.64, so about 36% of the variance of the jet fuel cost remains after hedging. That the R^2 equals the hedge ratio here is a coincidence of the inputs: $R^2 = \rho^2$ while $h^* = \rho\,\sigma_S/\sigma_F$, and the two agree only because $\sigma_S/\sigma_F = 0.8$ happens to equal ρ.

Estimating the hedge ratio in practice

Equation (3.3) requires three numbers that must be estimated, and the estimation choices matter more than the formula.

Use changes, not levels. Regressing the spot price on the futures price rather than their *changes* typically produces an R^2 near one, and a hedge ratio near one whenever the two levels share a trend of similar slope. Both series are nonstationary and trend together; the regression is measuring that shared trend, not the co-movement a hedge relies on. (Cointegration, a long-run tie between the two levels, is a separate question and does not change the point.) This is among the most common errors in applied hedging, and it flatters the result badly.

Match the data frequency to the hedge horizon. A hedge held for a month should be estimated on monthly changes. Daily data offers more observations and a tighter standard error, but it measures a different thing: short-horizon co-movement is often weaker than long-horizon co-movement, because temporary dislocations between spot and futures wash out over time. Example 3.4 shows how large the difference can be.

Example 3.4 The hedge ratio depends on the horizon

The same spot and futures series are used to estimate (3.3) at two frequencies.

Data frequency	σ_S	σ_F	ρ	h^*
Daily changes	0.0070	0.0094	0.62	0.46
Monthly changes	0.0320	0.0400	0.80	0.64

The two estimates differ by nearly 40%. The daily figure is not wrong; it is the right hedge ratio for a hedge held one day. Applying it to a one-month hedge underhedges substantially, because it has been calibrated on a horizon over which spot and futures diverge more.

Most of the rise comes from the correlation, which climbs from 0.62 to 0.80; the rest comes from the volatility ratio σ_S/σ_F, which edges up from 0.745 to 0.800. The general rule is that correlation between related assets usually rises with the measurement horizon, so the hedge ratio does too. A hedger who has a monthly exposure and a daily dataset should aggregate the data, not the reasoning.

The sample period is a judgment, not a technicality. A long sample gives precision; a short one gives relevance. The minimum-variance ratio is not a constant of nature: it shifts when market structure changes, when a new delivery point opens, when refining capacity changes the relationship between two grades. A ratio estimated over ten years spanning a structural break is precisely estimated and wrong.

Overlapping observations. Using monthly changes measured weekly, a common device for increasing sample size, creates overlapping windows and correlated residuals. The point estimate is usually acceptable; the standard errors are not, and a hedger who concludes the ratio is precisely determined on this basis is mistaken.

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Tailing the hedge. Equation (3.4) works in units, and it slightly overhedges. Because futures are settled daily, a gain realized today earns interest until the hedge is lifted, so a smaller position delivers the same offset at the horizon. Discounting over the remaining hedge life $T - t$ at the interest rate r gives

$$N^* = \frac{h^* Q_A}{Q_F} e^{-r(T-t)} \tag{3.5}$$

a correction known as **tailing the hedge**; [TN3] derives it and tabulates the factor. The correction in (3.5) is small for short horizons and grows with maturity and with the level of interest rates: at $r = 4\%$ it removes 0.3% of the position over one month but 7.7% over two years, and the position is scaled back up as the horizon shortens. It comes from daily settlement, as the forward–futures price difference of Section 5.12 does, though unlike that difference it does not require interest rates to be uncertain.

Ratios estimated on returns. Futures hedge ratios are more often estimated on *proportional* changes than on absolute ones (returns on the spot against returns on the futures), which is the natural scale when the position is resized as prices move. Write $\hat{h}$ for a ratio estimated that way, and let S and F be the spot and futures prices when the hedge is placed (the S_1 and F_1 of Section 3.3, with the subscript dropped because only one date is in play here). Then $\hat{h}$ is not the h^* of (3.3). A proportional change becomes an absolute one by multiplying by the price, so $\sigma_{\Delta S/S} = \sigma_S/S$ and $\sigma_{\Delta F/F} = \sigma_F/F$, while ρ is unchanged because correlation does not depend on scale. Hence

$$\hat{h} = \rho \frac{\sigma_S/S}{\sigma_F/F} = h^* \frac{F}{S}.$$

A ratio estimated on returns is therefore applied to *values* rather than quantities,

$$N^* = \frac{\hat{h} V_A}{V_F} \tag{3.6}$$

where $V_A = SQ_A$ is the value of the position being hedged and $V_F = FQ_F$ the value of one contract. Substituting takes (3.6) straight back to (3.4):

$$\frac{\hat{h} V_A}{V_F} = h^* \frac{F}{S} \cdot \frac{SQ_A}{FQ_F} = \frac{h^* Q_A}{Q_F}.$$

The two are the same hedge written two ways, and either is correct on its own terms. What is not correct is to mix them: an h^* estimated on price changes and then applied to values gives

$$\frac{h^* V_A}{V_F} = \frac{h^* Q_A}{Q_F} \cdot \frac{S}{F},$$

rescaling the position by S/F, shrinking the hedge in contango, when futures prices stand above spot and rise with maturity, and enlarging it in backwardation, when they stand below spot and fall with maturity, for no reason at all.

3.6 HEDGING AN EQUITY PORTFOLIO

Stock index futures let a manager change a portfolio's market exposure without trading its constituents. This is the use Chapter 6 identifies for Treasury futures and duration.

The relevant hedge ratio is the portfolio's **beta**, its sensitivity to the index. To hedge fully,

$$N^* = \beta \frac{V_A}{V_F} \tag{3.7}$$

where V_A is the portfolio's value and V_F the value of one futures contract.

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Example 3.5 Hedging a portfolio

A manager holds a \$5,050,000 portfolio with a beta of 1.5 against an index standing at 4,200. Futures contracts are worth \$50 per index point, so each contract is worth $4{,}200 \times 50 = \$210{,}000$. From (3.7),

$$N^* = 1.5 \times \frac{5{,}050{,}000}{210{,}000} = 36.07$$

so the manager shorts 36 contracts.

Suppose the index then falls 10% to 3,780. The gain on the futures is

$$36 \times (4{,}200 - 3{,}780) \times 50 = \$756{,}000$$

while the portfolio, with a beta of 1.5, falls by roughly 15%, or \$757,500. The two offset to within \$1,500.

Removing the market also removes the market's return. Setting aside dividends and the gap between futures and spot prices, a fully hedged portfolio earns roughly the risk-free rate plus whatever the stocks earn beyond what their beta predicts (their alpha).

Changing beta rather than eliminating it. A manager who wants to reduce exposure without removing it shorts

$$N^* = (\beta - \beta^*)\,\frac{V_A}{V_F} \tag{3.8}$$

contracts, where β^* is the target. In Example 3.5, where $V_A/V_F = 24.05$, equation (3.8) moves beta from 1.5 to 0.75 by shorting $0.75 \times 24.05 = 18.04$, so 18 contracts; *raising* beta to 2.0 gives a negative count, and the manager goes long $0.5 \times 24.05 = 12.02$, so 12.

Why hedge rather than sell? If the manager wanted no market exposure, selling the portfolio would achieve it. Futures are used instead for three reasons: the manager may believe the specific stocks will outperform and want to keep that bet while removing the market, transaction costs on index futures are far lower than on the underlying holdings, and the position may be intended as temporary. The first of these is the most important: hedging with index futures is how a manager separates stock selection from market timing.

Hedging is not a one-off decision

Equations (3.4) and (3.7) give a position to put on today. Both quantities they depend on then move.

The portfolio's value changes, so the hedge sized against V_A becomes wrong; the beta changes as holdings change and as the market's factor structure shifts; and for a commodity hedger the exposure itself changes as production forecasts are revised. A hedge placed once and left is correct only on the day it was placed.

How fast does it drift? For the equity hedge the answer is available in closed form. If the index moves by a proportion m, the portfolio moves by βm and one contract by m, so the position required by (3.7) changes by the factor

$$\frac{1 + \beta m}{1 + m}.$$

Two things follow. At $\beta = 1$ the factor is exactly one: market moves alone never call for rebalancing, because the exposure and the contract are revalued in step. What a manager is chasing is the amount by which β differs from one. And even away from $\beta = 1$ the drift is slow.

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In Example 3.5 the 10% fall leaves a portfolio of $4,292,500 and a contract worth $189,000, so the position now required is

$$1.5 \times \frac{4{,}292{,}500}{189{,}000} = 34.07$$

against the 36 contracts held, a two-contract adjustment after a 10% move in the market.

Rebalancing has a cost (each adjustment crosses a spread), so in practice firms use a schedule and a trigger together: monthly for an equity overlay (a futures position laid over a portfolio that is itself left untouched), on the planning cycle for a commodity hedger whose forecast exposure is the thing that moves, with a band of a few percent of the required position to catch large moves in between and a forced adjustment at every roll. The ratio itself is re-estimated far less often than the position is resized, quarterly at most, because re-estimating on each new observation feeds sampling noise straight into the position, and [TN3] puts a number on that cost. The choice of band is the same trade-off that recurs throughout this book: continuous adjustment is accurate and expensive, infrequent adjustment is cheap and approximate.

This is also the first appearance of an idea that dominates Part III. Chapter 18 shows that hedging an option requires exactly this kind of continuous rebalancing, because an option's sensitivity to the underlying changes as the underlying moves. The difference is one of degree: here the hedge ratio drifts slowly, and there it moves whenever the market does.

3.7 HOW MUCH TO HEDGE

Equation (3.3) answers a narrow question: what ratio minimizes the variance of a single exposure over a single horizon? Corporate hedging programs rarely answer only that question, and it is worth seeing what else enters.

Partial hedging is normal and often deliberate. Many firms hedge a fixed fraction of expected exposure (half the next year's production, a quarter of the year after) rather than the variance-minimizing amount. Three reasons follow, all defensible:

The exposure itself is uncertain. A producer hedging 100% of *expected* output is overhedged if output disappoints, and then holds a naked short futures position that loses if prices rise, which an industry-wide shortfall tends to make them do. Hedging a fraction leaves room for the quantity to come in low.

Shareholders may want some exposure. An investor buying a copper miner is buying copper exposure. A fully hedged miner is a fabrication business with a mine attached, which is not what was bought.

Cash constraints bind. Section 3.10 is about this, and it is the reason a variance-minimizing ratio can be the wrong answer.

Layered hedging. A common structure hedges a declining fraction across a rolling horizon (perhaps 75% of the next six months, 50% of the following six, 25% of the year after), adding a new layer each quarter. This averages the entry price across many dates, which removes the risk of having placed the entire program on a single bad day, and it degrades gracefully if forecasts change.

The point is that the entry price is itself a risk. A firm that hedges its full three-year exposure in one transaction has replaced price risk with timing risk, and Case Study 3.1's organizational problem then attaches to a single decision made on a single day.

Or hedge with options instead. Futures are not the only instrument. A firm that buys an **option** (a producer a put, a consumer a call) removes the unfavorable tail and keeps the favorable

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one, in exchange for a premium. A purchased option whose premium is paid up front then owes no variation margin at all. That difference bears directly on the failures of Section 3.10, but options are not developed until Chapter 10, so the comparison is made in Section 12.10.

3.8 ACCOUNTING FOR HEDGES

Case Study 3.1 described the organizational difficulty of hedging. There is a formal version of it in the accounts, and it shapes which hedges firms actually choose.

Derivatives are carried at fair value with changes recognized in profit or loss. The hedged item usually is not: inventory sits at cost, future production is not on the balance sheet at all, and a forecast purchase does not exist as an accounting item. So a perfectly effective economic hedge produces *earnings volatility from the derivative alone*, with nothing offsetting it until the hedged transaction occurs.

Hedge accounting exists to fix this, in one of two ways depending on what is hedged. In a **fair value hedge**, of an asset or liability already on the balance sheet, the hedged item is remeasured for the hedged risk and the change goes to profit or loss beside the derivative's, so nothing is deferred; inventory hedged with futures, the case above, is of this kind. In a **cash flow hedge**, of a forecast transaction, there is no item to remeasure, so the effective portion of the derivative's gain or loss is held in other comprehensive income, a component of equity outside profit, and moved into profit when the hedged transaction affects earnings. Either way the two legs meet in the same period, which is what the economics always said.

The requirements are not trivial. Documentation must exist at inception, and the designation constrains what the firm may subsequently do with the position. The effectiveness test differs between the two main standards. International Financial Reporting Standard (IFRS) 9 requires an economic relationship between the hedged item and the hedging instrument, assessed prospectively, with any ineffectiveness recognized in profit or loss as it arises; it dropped the retrospective 80–125% test of its predecessor. US generally accepted accounting principles (US GAAP), in Accounting Standards Codification (ASC) Topic 815, still require the hedge to be highly effective, but after inception allow that to be assessed qualitatively.

The consequence for hedging choices. Because qualifying is onerous and failing is visible, firms sometimes prefer a hedge that qualifies for hedge accounting over one that is economically better. IFRS 9 eases one version of the problem by allowing a risk component to be designated as the hedged item, such as the crude oil component of a jet fuel purchase, so a crude hedge of jet fuel can qualify for that component. But a cross hedge whose link to the hedged item is real yet hard to evidence may still fail to qualify, or qualify with large recorded ineffectiveness, where a worse-matched but simpler instrument passes. A dynamic strategy that adjusts as exposures change may be harder to document than a static one that hedges less well.

This is worth stating plainly: the accounting treatment is a real input into hedging decisions, and it does not always point the same way as the economics. A reader who finds a firm's hedging program puzzling should check whether the puzzle dissolves once the accounting constraint is added.

And the tax rule is a third answer

Tax authorities also have a view, and it is not the accounting one. The specifics are jurisdictional and change, so what follows is the shape of the problem rather than any country's current rules; a reader acting on it should check the rules in force.

Three features recur across jurisdictions and matter to a hedger. *Timing*: gains on an exchange-traded futures position are commonly taxed as they accrue (marked to market at the year end whether or not the position was closed), while the hedged item is taxed when it is

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sold. *Character*: the derivative's gain may be a capital gain while the hedged item produces ordinary income, and capital losses often cannot be set against ordinary income. *Qualification*: some systems, notably the United States, provide a hedging exception that aligns both, taxing the derivative on the same timing and character as the item it hedges, but it must be claimed by identifying the hedge on the day it is entered into; others largely follow the accounting treatment.

The failure mode is the one this chapter keeps meeting. A hedge that is economically exact can be taxed asymmetrically: the loss leg deductible in one year at one rate, the gain leg taxable in another at another. The firm is then paying tax on a profit it did not make, in a year in which its economic position did not change—and it is paying cash, on the schedule of Section 3.10's margin calls rather than on the schedule of its production. Metallgesellschaft, whose collapse Section 3.10 describes, met the same asymmetry in its accounts rather than its tax bill: German accounting rules, which recognize unrealized losses but not unrealized gains, booked the losses on its futures while ignoring the offsetting gains on its customer contracts.

So a hedging decision faces three tests and they do not agree. The economics ask whether the exposure is offset, hedge accounting asks whether the offset can be documented and demonstrated, and the tax rule asks whether the hedge was identified in time and in the right form. *A hedge can pass any two and fail the third*, and the firm discovers which one it failed only afterward.

3.9 ROLLING THE HEDGE FORWARD

Section 3.3 assumed a contract exists spanning the exposure. Often none does. Liquidity in futures concentrates in the nearest few contracts, while a firm's exposure may extend years.

The response is to **stack and roll**: place the entire exposure in a near contract, and as it approaches expiration, close it and open the equivalent position in the next. A five-year exposure hedged with quarterly contracts requires nineteen rolls.

Each roll introduces basis risk of its own: the hedger closes at one contract's price and opens at another's, and the spread between them is not known in advance. Nineteen rolls means nineteen draws from that distribution. Example 3.6 works one sequence of rolls through, and Example 3.7 shows what the draws can cost when they run against the hedger.

Example 3.6 A rolling hedge, roll by roll

A producer will sell 1 million barrels in twelve months, but only three-month contracts are liquid. It shorts the near contract and rolls quarterly. The market is in mild contango throughout.

Month	Action	Price	Gain on leg
0	Short contract A	80.00	
3	Close A	78.00	+2.00
	Short contract B	78.60	
6	Close B	76.50	+2.10
	Short contract C	77.00	
9	Close C	75.00	+2.00
	Short contract D	75.40	
12	Close D	74.00	+1.40
	Sell the oil at spot	73.90	
	Total futures gain		**+7.50**

The effective price is 73.90 + 7.50 = \$81.40 per barrel—\$1.40 *better* than the \$80.00 at which the hedge was placed.

The improvement decomposes cleanly. Each roll closed a contract and opened a more

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expensive one: +0.60, +0.50, +0.40, totaling \$1.50 of roll gains. The final basis contributed 73.90 − 74.00 = −\$0.10. Together, 80.00 + 1.50 − 0.10 = \$81.40.

The producer gained because it was *short* in a market in contango. Reverse the direction (a consumer hedging with long futures in the same market) and the identical sequence of prices produces \$1.50 of roll *losses*.

Example 3.7 The cost of rolling

A producer must hedge 1 million barrels a month for three years, but only the front three contracts are liquid. It stacks the entire remaining exposure in the front contract and rolls monthly: 35 rolls, since the last month's barrels are delivered from the front contract without a further roll.

Suppose each roll costs 15 cents a barrel, because the market is in backwardation, which works against a short hedger. The position rolled shrinks from 35 million barrels to 1 million, an average of 18 million,

$$35 \text{ rolls} \times \$0.15 \times 18{,}000{,}000 = \$94.5 \text{ million}$$

of cumulative roll cost, against a hedge that was placed to protect a price.

Example 3.6 showed which direction the cost runs: a short hedger *gains* from rolling in contango and loses in backwardation, and a long hedger the reverse (Section 5.11 works the long investor's case). The producer above is short in backwardation and so pays. The mirror case, a firm hedging the other way with long futures, faces exactly this bill when the market moves into contango.

That is not a hypothetical. It is what happened to Metallgesellschaft.

3.10 WHEN HEDGING FAILS

The mechanisms above (basis risk, roll risk, and the cash flow timing of Chapters 2 and 8) combine badly. The clearest illustration remains a case from the early 1990s, told in Case Study 3.2.

Case Study 3.2 Metallgesellschaft

The US subsidiary of the German firm Metallgesellschaft sold long-term contracts to deliver heating oil and gasoline at fixed prices, some running as long as ten years. It had committed to sell roughly 160 million barrels at prices fixed in advance.

That is a short physical position, and it was hedged by going *long* futures. Since no liquid ten-year contract existed, the firm stacked the exposure in near-dated futures, about a third of the position, and short-dated over-the-counter swaps, and rolled them.

Three things then happened at once. Oil prices fell sharply through 1993, so the long futures position lost money—and because futures are settled daily, those losses had to be paid in cash immediately. The market shifted from backwardation into contango, so each roll now cost money rather than earning it. And the offsetting gains, sitting in customer contracts running out to a decade, produced no cash at all.

The margin calls ran to roughly \$900 million by the end of 1993, and the losses eventually came to about \$1.3 billion. The parent, MG AG, was itself in difficulty and was rescued by its banks in early 1994. Facing a cash demand of that size against gains it could not realize, it chose not to fund its subsidiary's margin: it liquidated the futures positions and unwound many of the customer contracts, crystallizing the losses while abandoning the position that would have offset them.

On the reading of Culp and Miller (1995) the hedge largely held in value, the futures losses being matched by gains on the supply contracts, and what it lacked was the cash

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> to carry that offset through time. Mello and Parsons (1995) and Edwards and Canter (1995) disagree: a stack of short-dated contracts sized one for one against commitments a decade long was not a sound hedge in the first place. Whether the parent was right to liquidate rather than fund the position remains debated. What is not in dispute is the mechanism, and it is the one this book keeps meeting: a position hedged against price risk, defeated by the timing of the cash it demanded, and abandoned at the point of maximum loss because the combined position was judged one leg at a time.

The 2022 episodes of Chapters 2 and 8 are the same story with different commodities and a different resolution. European utilities that had sold output forward faced margin calls in the tens of billions against physical gains years away. The difference in outcome was not that those firms hedged better; it was that governments extended credit lines to them, while Metallgesellschaft's parent chose not to fund its subsidiary's margin.

What to take from this. Three practical conclusions follow, and they are the reason this chapter exists in the form it does.

Size the hedge to the cash you can produce, not only to the exposure. A hedge ratio derived from (3.3) minimizes price variance and says nothing about liquidity. A firm that cannot fund the margin on its variance-minimizing hedge should hedge less—and should know that in advance rather than discovering it. Often the cash problem is solved by a different instrument rather than a smaller hedge: the reserve-based borrowers of Section 3.2 usually hedge with swaps from their lending banks, secured by the same collateral as the loan and carrying no variation margin.

Stack and roll converts a long-dated exposure into a sequence of short-dated ones, and the sequence has its own risk. It is not a substitute for a matched-maturity hedge; it is a different strategy with different failure modes.

Judge the combined position. Every failure in this section involved someone looking at the futures leg alone.

SUMMARY

A short hedge protects a firm that owns or will own an asset; a long hedge protects one that must buy. Whether to hedge at all is a real question, since shareholders can diversify; the stronger arguments concern frictions, above all the cost of financial distress. Hedges are also hard to sustain, because the loss on the futures leg is visible while the offsetting gain is buried in operating results, so a program must be mandated in advance and judged on the combined position.

A hedge lifted before delivery locks in $F_1 + b_2$, so the risk that remains is the basis at the close, which arises from mismatches of asset, location, and timing. Basis risk is at least as important in financial hedges and less visible: a corporate bond portfolio hedged with Treasury futures keeps its credit spread risk, and a hedge in a correlated proxy currency fails exactly when the two currencies part company.

The variance-minimizing ratio $h^* = \rho\,\sigma_S/\sigma_F$ is a regression slope, and its R^2 says how much of the risk the hedge removes. It must be estimated on changes rather than levels and at the frequency of the hedge horizon. An equity portfolio is hedged with index futures in proportion to its beta, which separates stock selection from market exposure.

Firms rarely hedge the full variance-minimizing amount, and purchased options whose premium is paid up front, which then owe no variation margin, are the alternative that avoids the funding failures. Accounting and tax can make an economically sound hedge look bad: the derivative is marked to market while the hedged item often is not, hedge accounting realigns

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the two only for hedges that qualify, and a tax rule may treat the two legs differently. Exposures longer than the liquid contracts are stacked and rolled, which adds a fresh basis at every roll. Metallgesellschaft combined every failure mode, and the 2022 energy episodes repeated the mechanism; what separated the outcomes was whether someone would fund the margin, not better hedging.

Companion notebook `ch03-hedging-strategies.ipynb`

FURTHER READING

Ederington (1979) is the origin of the minimum-variance hedge ratio in the form this chapter uses, and it is short. Froot, Scharfstein, and Stein (1993) make the modern case for why a firm hedges at all—the question behind every hedge ratio and the one a formula cannot answer. On Metallgesellschaft the literature is a genuine disagreement and should be read from both sides: Culp and Miller (1995) argue the strategy was sound and the liquidation was the error, while Edwards and Canter (1995) and Mello and Parsons (1995) take the opposing view. Smith and Stulz (1985) set out the tax and distress-cost arguments for hedging that Froot, Scharfstein, and Stein build on.

PRACTICE QUESTIONS

3.1 Set out the argument that a firm should not hedge because its shareholders can diversify, and give two reasons the argument may fail.

3.2 Define the basis, show that a short hedger who lifts a hedge before delivery receives $F_1 + b_2$, and explain what this implies about where the residual risk lies.

3.3 A hedge is placed when the spot price is \$44 and the futures price is \$41.50, and lifted when the spot price is \$47 and the futures price is \$45.20. Compute the effective price for a short hedger, state whether the basis strengthened or weakened, and explain why a June exposure is usually better hedged with a July contract than a June one.

3.4 A company will buy 800,000 units of a commodity in three months. The standard deviation of the change in its price over three months is \$0.65, that of the futures price \$0.81, and the correlation 0.72. Each contract covers 50,000 units. Compute the minimum-variance hedge ratio and the number of contracts, state what proportion of the variance remains, and explain why that ratio is the slope of a regression.

3.5 A \$12 million portfolio has a beta of 0.9. Index futures are worth \$50 per point and the index stands at 4,000. How many contracts hedge it fully, and how many reduce beta to 0.4?

3.6 An analyst estimates a hedge ratio of 0.46 from daily data and 0.64 from monthly data for a hedge held one month. State which should be used, explain why the two differ in this direction, and explain what goes wrong if the regression is run on levels rather than changes.

3.7 Give three reasons a firm might deliberately hedge less than the variance-minimizing amount, and explain what layered hedging achieves that a single transaction does not.

3.8 A manager hedges an equity portfolio with index futures using the beta-adjusted contract count. Show that if the index moves by a proportion m, the position required changes by the factor $(1 + \beta m)/(1 + m)$. Explain why a portfolio with a beta of one never needs rebalancing on account of a market move alone, state the condition on β and m under which the hedge must be *increased*, and evaluate the factor for a portfolio with a beta of 0.8 after the index rises 12%.

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3.9 A bank hedges a portfolio of corporate bonds with Treasury futures. Explain what basis risk remains, why a historical regression over a calm period would understate it, and what happens to both legs in a credit sell-off.

3.10 Using the roll table of Example 3.6, recompute the effective price if the market had been in backwardation, with each roll closing at a price 0.40 *above* the next contract, and state who gains in each case.

FURTHER QUESTIONS

3.11 A fund manager wishes to keep a portfolio's stock selection but remove its market exposure for three months. Explain why index futures are preferred to selling the portfolio, and identify what risk remains after the hedge is in place.

3.12 Explain what stack and roll is, why a firm would use it, and the two distinct risks it introduces that a matched-maturity hedge would not. Relate the direction of the roll cost to the shape of the forward curve.

3.13 Set out the Metallgesellschaft episode as a sequence of events, identifying at each stage which of this chapter's risks was operating. Was the strategy unsound, or was the liquidation the error? Defend a position.

3.14 A producer's variance-minimizing hedge would require it to post margin that could reach 40% of its annual revenue in a severe price move. Explain why the hedge ratio of equation (3.3) does not capture this, what the firm should do, and how you would set the hedge ratio instead.

3.15 A firm chooses a hedging instrument that matches its exposure less well than an available alternative, on the grounds that the alternative would not qualify for hedge accounting. Set out the argument for and against that decision, and say what you would need to know about the firm to judge it.

3.16 European utilities in 2022 and Metallgesellschaft in 1993 faced the same mechanism with different outcomes. Identify the mechanism precisely, explain what differed, and state what a treasurer should arrange in advance so that the difference does not matter.

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